

Before the starting the design check the unconstrained paths in sanity check level only..

```
innovus -log Log/floorplan.log          -- To save the log files in a
perticlry floder.                      --
dbGet head.rules.name                  -- To know on which type of rule
will work on the design. It will displays all the rules.
report_design                          -- It will shows the Design info
and Design rules..
```

```
sizeof_collection [all_fanout -from clk] -- To findout which clk is going to
whitch flop.
```

```
sizeof_collection [get_cells -hierarchical] -- To find out the lib cells
count in our design.
```

```
sizeof_collection [all_registers -flops] -- To know all filp flops in our
design.
```

```
sizeof_collection [all registers -latches] -- To know all latches in our
design
```

```
llength [dbGet top.Insts.cell.name *BUF*] -- To count the BUF in our design.
```

```
sizeof_collection [get_lib_cells -of_objects [get_cells]] --To findout the lib
cells count.
```

```
sizeof_collection [get_cells] -- To get all cells in our design.
```

```
get_cells -hierarchical -filter "is_black_box == true" --To findout the macro
in our Design.
```

```
get_pins -of_objects [get_cells out_reg_0] -- Get pins of that
particular cell.
```

```
get_pins -of_objects [get_cells out_reg_0] -filter "direction==in" --To findout
the cell name by its pins Input/Outpins.
```

```
check_instance_library_in_views -- Checks for missing
libraries for instances in the active view.
```

```
get_time_unti -- It will displays the time units.
```

```
get_capacitance_unit -- It will displays the capcitanace
units.
```

```
get_resistan_unit -- It will displays the resistance units.
```

```
get_clocks -- It will displays the clock names in
```

```
our design.
```

```
get_lib_cells * -- To get lib cells in our design.
```

```
get_lib_pins * -- To get lib cells along with lib
cells.
```

```
checkPinAssignment --
```

```
get_property [get_pins cells/CK] clocks -- To findout the either that
pin is getting clk or not.[it will displays empty that means no clk]
```

```
get_property [get_pins some cell names] clocks --To know its pin
connections..
```

```
setAnalysisMode -checkType setup/hold -- to chenge the timing
report mode.
```

```
set_interactive_constraint_modes [all_constraint_modes -active]
```

```
all_analysis_views
```

```
set_analysis_view -setup scan_shift_max -hold scan_shift_min
```

```

floorplan & Powerplan commands
reportDensityMap --
reportPinDensityMap --
checkFPlan -- It should be done after placing the
pins.And it will displays the some date
checkFPlan -reportUtil -- To know the utilization on each
stages.
setNanoRouteMode -routeTopRoutingLayer 3 -routeBottomRoutingLayer 1 --That means
the routing layers which are used in our design upto 3 only.
verifyConnectivity -type all -error 1000 -warning 50 -- To verify the connections.
verifyGeometry -- It will displays the cells
placemnets and adjustments.
verify_PG_short -- For verifing vdd and vss power
shorts..
verify_drc -- For verifing the drc checks

```

```

set inti_pwr_net VDD
set inti_gnd_net VSS
globalNetconnect VDD -pin VDD* -type pgpin
globalNetconnect VSS -pin VSS -type pgpin

```

Give the setDontuse CLKBUF\* and CLKINV\* true before going to placement  
 After Placement commands  
 (Will check only Setup check in placement not hold.. Hold will be done at after CTS stage)

```

report_constraint -drv_violation_type max_transition -- To know
the Max_tran violations values.
report_constraint -drv_violation_type max_transition Some cell name/with pin -- To
know tran violation by each cell.

```

(Ex

```

dma_dut/dmamaster_dut/FE_OCPC48_hwddata_m_0/A)

```

```

place_opt_design -- For std cells placement and preCTS
optimization..
report_ports -- Will displays the all ports like
in and out with pin names and direction and rise and fall tym also..
checkPlace -- It will displays as placed and
unplaced cells info and Placement Density.
Check_timing -- It will displays the all info
about Timing related..Max it should be cleared
report_clocks -- It will displays the clocks info
which are presnet in our design. like main clk, and generated clk,V clk..
report_constriants -all_violaters -- The constraints information
includes information such as whether or not constraint is violated,
by how much amount, and the worst
violator object.

```

It will displays the DRC

```

values(max_tras, max_cap, max_fanout), If the DRC are not MET then only will
get DRV.(otherwise will not
displayed).
reportCongestion -overflow -- It will shows the how much area
is congested.
report_timing_derate --
A.report_constraint -all_violators -- To see all the vilated path in
our design (setup vilations).
B.report_timing -max_slack 0 -nworst n(10) -- It will shows upto 10
numbers of violated paths. (both commands are same report_constraints and this
command).
C.report_timing -max_paths n (10) -- It will shows total how many
paths are available in our design..It will displays the MET and Vilotaed paths.
D.report_timing -begin_end_pair -- Same as above command
(A,B,C,D).
report_timing -check_type setup -nworst 100 -- Reports all the path which
are setup and hold violatoins..
report_timing -unconstrained -nworst 100 -- It will displays all the
unconstrained paths in our design.(After doing this step pick one endpoint cell pin
and check its getting clk or not by
using this get_property [get_pins any cell pin name] clocks
reportAlwaysOnBuffer -all -- It will shows the all low power
ON buffers in our design.
getAnalysisMode -check all types -- It will gives the on what type
of mode we are working and etc..
report_annotated_check -- It will displays the arc
setup,recovry,min pulse width,ect...

setOptMode -fixFanoutLoad true -- To reduese the fanout
violations.
set_max_fanout 10 -- To set the fanout limite.

set_ccopt_mode -cts_opt_type cluster|tril|full [select cluster]
ccopt_design

```

Give the setDontuse CLKBUF\* and CLKINV\* false before going to cts

After CTS

```

get_propagated_clock -clock {*} -- To know the clocks positon
either it is in ideal or propagated position.

```

```

report_clock_gating_check -- It will
displays the all clock getting points.
get_property [get_cells pcm_inst/pwr_clock_gate/g38] is_clock_gating_check -- To
check the either clock_gating cell or not.
get_property [get_cells pcm_inst/pwr_clock_gate/g38] ref_name
Some Gates name (like CLKAND gates) -- To findout its
lib name.
get_property [get_cells pcm_inst/pwr_clock_gate/g38] is_combinational
true -- That means its a
combinational cells so we can Disable it.

```

```

place_opt_design -- Executes pre-CTS flow with both
placement and pre-CTS optimization.
    Earlier, users had to perform the
    following two steps to do placement and preCTS optimization:
    * placeDesign * optDesign -preCTS
    With place_opt_design, users need to
run only a single command to do placement and preCTS optimization.
    The place_opt_design command provides
better integration between placement and optimization to achieve
    faster runtime and better PPA.

timeDesign -prePlace -- Ignores the net load while
building the timing graph and runs timing analysis for generating timing reports
    before the design is placed
timeDesign -postRoute -- run a postRoute parasitic
extraction (TQuantus by default), and timing analysis to generate timing reports
for
    both setup and hold
violations after the design has been routed.
report_clock_timing -type latency -- To check the latency value
either violated or not.(max tool will fix the latency value)

```

```

After Routing
verify_PG_short -- It will displays the PG short in
our design.
routeDesign -- To route the disegn..
reportCongestion -overflow -- To check the Congestion by
overall design...
report_constraint -all_violators -- To check the violations..
timeDesign -postRoute -- To run this command we need to
set the analysis mode for that use this command
(setAnalysisMode -analysisType
onChipVariation)
optDesign -postRoute -drv -- If there is any violations in
our design then use this command for clearing the violations...
    It will decreas the violations. that
means optimize the errors.
optDesign -postRoute -hold -- For hold violations..
verify_drc -- To verify the
globalNetConnect VDD -type pgpin -pin VDD -- To clear the short violation in
VDD
globalNetConnect VSS -type pgpin -pin VSS -- To clear the short violation in
VSS

```

```
report_constraint
    - check_type
    - all_violators
    - late -all_violators
    - check_type recovery -all_violators
```

If you get any hold violations then use the buffers (repeaters)  
get\_nets -of\_object any Instance name(Ex IRReg/q\_c\_reg\_2) -- To get net names like  
where we can add the buffers  
ecoAddRepeater -net q\_c[2] -cell BUFX3 -relativeDistToSink 1 -- It will add the  
buffer.

Extraction:-  
create one directory  
mkdir any file name(Ex finalout)

create Netlist  
saveNetlist finalout/netlist.v

create def  
defOut finalout/counter.def

create spef  
setExtractRCMode -effortlevel low -engine postRoute  
extractRC  
rcOut -spef finalout/counter.spef

Global net connections are to remove the Shorts and Opens..

for fixing the max\_tran violation with tcl  
1.copy the violations in a file.  
2.put numbers by line by line.  
3.After putting the number copy the line from where to where u want to copy (sed -n  
'15,132p' max\_tra\_vios)  
4.then using awk print only coloms. (sed -n '15,132p' max\_tra\_vios | awk '{print  
\$2}')  
5.and copy the data in a file (sed -n '15,132p' max\_tra\_vios | awk '{print \$2}' >  
max\_tran\_less)  
6.last add the buffer (sed -n '1,60p' max\_tran\_less\_less | awk '{print

```
"ecoAddRepeater -cell BUFX3 -term "$1""}')
```

```
selectInstByCellName CLK*
```

```
llength [dbGet selected.name ]      --To findout clk cells in data path or  
not(before placment) will count comes then it means RTL ppl will allow those cells.
```

```
set k [all_fanout -from ethernet_mac_2/rx_fifo_module_to_dma/g5413/Y]
```

```
foreach s $k {
```

```
+ puts [get_object_name [get_lib_cells -of_objects [get_object_name [get_cells  
-of_objects [get_pins $s]]]]]
```

```
+ }
```

```
To finout the
```

```
lib cells of a perticular instnce
```

```
get_lib_cells -of_objects ethernet_mac_2/rx_fifo_module_to_dma/g5413
```